# Data-Centric AI: Comparing Lap Analysis from Raw versus Structured F1 Telemetry

Piastri · Japan · 2026 (single-lap samples)

**Abstract**

This note compares two analyses of materially the same single-lap telemetry: one confined to raw merged exports, one augmented with project-defined engineered columns. The downstream analytic protocol (persona, refusal to invent session facts) is held similar; the controlled variable is *tabular representation*—in the spirit of data-centric AI, not "model ingenuity."

## 1 Purpose and framing

This report compares two analyses of what is materially the same single-lap telemetry—one confined to the raw export `piastri_2026_japan_fastest_lap_telemetry.csv` (**raw**), one using `piastri_2026_japan_structured.csv`, which adds heuristic engineering columns: `Corner_Type`, `Driving_State`, and distance-based deciles labelled `Mini_Sector`. The downstream analytic tool (prompt and LLM persona) was intentionally similar: evidence-grounded narration, refusal to infer official lap or sector timing, tyre compound, weather, or other facts absent from the CSVs. The **controlled variable**, in a **data-centric AI** framing, is therefore **feature and table design**—not a change of foundation-model family.

## 2 Dataset differences beyond column count

**Raw telemetry.** The raw file exposes `Speed`, throttle, brake, cumulative `Distance`, merge metadata (`Source`), and coordinates. Meaningful segmentation requires explicit analytic choices—for example quartiles by row index (Q1–Q4 as a heuristic) and slow-speed clusters cited by metre along `Distance`. The narrative can *withhold* named corners because circuit geography is not keyed as text columns in this export.

**Structured telemetry.** The structured file **reuses identical sensor traces** yet relocates cognition from free-form slicing to preprocessing: categorical summaries of longitudinal intent (`Driving_State` counts), coarse corner-style buckets (`Corner_Type`), and ten distance bins (`Mini_Sector`), enabling tabular summaries per mini-sector instead of hand-built cuts.

## 3 Aligned empirical anchors

Across both summaries, headline statistics **align**: **Speed** spans roughly 73–317 km/h (FastF1-style magnitude); approximately **19%** of samples have `Brake=True`; mean `Throttle` $\approx$ 65.8; elapsed sample span near **93 s**; the **DRS** column reads as uniformly inactive in this excerpt. Alignment on these anchors supports consistent empirical grounding despite different narrative affordances.

# 4 Outcomes: affordances versus epistemic risk

## 4.1 What raw emphasizes

Raw-grounded prose excels at **calling out limits**: normalized `RelativeDistance` not reaching 1.0; potential lap-boundary artefacts (e.g., high throttle simultaneously with braking at the earliest rows); ambiguity between interpolation and genuine coasting. These caveats are intellectually conservative and reduce over-claiming about driver technique.

## 4.2 What structured emphasizes

Structured analysis offers **immediate segment readability**: histogram-style views tying `Driving_State` and heuristic corner labels to each `Mini_Sector`, enabling shorthand such as dominance of transitional phases or comparatively high mean speed in bands like `MS_8`–`MS_9` without manual binning repetition. Fluency rises because labels collapse repeated statistical work into categorical columns—but each label inherits upstream threshold choices; readers may mistakenly equate heuristic tags with circuit truth or timing authority unless caveats persist.

# 5 Data-centric takeaway

Improvement attributable to switching inputs stems chiefly from **upstream representation**: engineered columns compress repeatable rules into shorthand that the same analytic scaffold can cite directly. That productivity gain is **not** immunity from telemetry limits (boolean brake, sparse merge), nor does it embed lateral tyre physics absent from columns.

# 6 Limitations statement

Neither table supports rival deltas, stint strategy, or optimised compound selection without additional data. Robust reporting should cross-check aggregates with a small scripted audit (ranges, percentages, grouped counts).

**Disclaimer.** Structured labels (`Corner_Type`, etc.) remain *project-defined*; distance minisectors are not FIA sector timers. Narrative flair must remain accountable to CSV evidence.